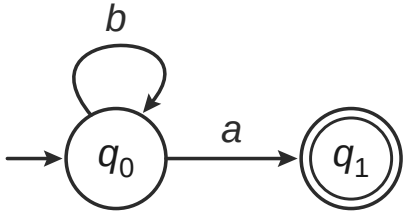
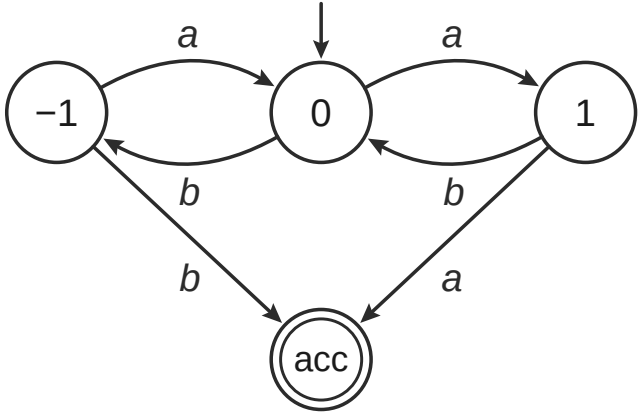
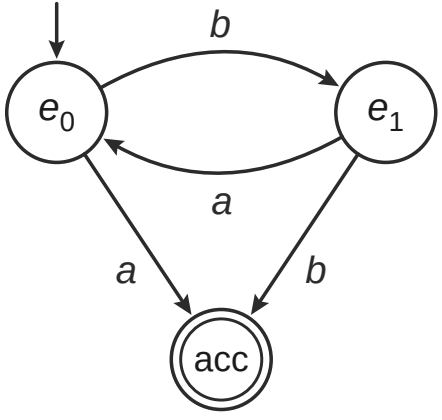
$\hat{P}_1 = b^* a$



$\hat{P}_2 = (ab+ba)^*(aa+bb)$



$\hat{P}_4 = (ba)^*(a+bb)$